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Building synthetic voices for under-resourced languages: the feasibility of using audiobook data

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***Abstract*—Creating synthetic voices that are natural and intelligible is a daunting challenge for well-resourced languages. The challenge is much bigger for languages in which the speech and text resources required for voice development are not available. Previous studies have suggested audiobooks as an alternative source of speech data. This paper reports on a comparison between voices derived from audiobook data and voices based on professional voice artist data. Two sets of voices were evaluated: male voices built using very small amounts of both data types (around 3 hours, representing a severely resource constrained scenario) and female voices trained on almost 10 hours of audiobook and professional speech data. The results of subjective listening tests indicate that, while the majority of the listeners preferred the voice artists' voices over the audiobook voices, the difference in naturalness was not perceived to be substantial. Results also showed that the artists' voices outperform the audiobook voices in terms of intelligibility, especially if a limited amount of training data is available. Although additional training data improves the intelligibility of audiobook voices, the results seem to indicate that a smaller quantity of professional data yields a better voice than large volumes of especially old audiobook data.**

***Index Terms*—speech synthesis, text-to-speech, under-resourced languages, audiobooks**

I. INTRODUCTION

Collecting and annotating the speech data required to develop a state-of-the-art text-to-speech (TTS) system requires a considerable amount of resources and expert knowledge. These requirements constitute a major challenge for under-resourced languages [1], [2]. Only a few languages have resources like the audio and text that are available through, for example, the Gutenberg project[1]. However, most countries do provide audiobooks to print-disabled readers through special libraries [3], [4]. The books are mostly read by volunteers, but the speech could still be used as an alternative source of data to develop synthetic voices for TTS systems, e.g. [5], [6].

Using audiobook data poses a number of challenges and limitations, especially if the language is under-resourced and the voices are to be distributed. Complicating factors could include *copyright* restrictions, *permission* to distribute voices, the availability of speech and text in *digital format*, the *pre-processing* required to prepare audiobook data for voice building as well as the development of appropriate *evaluation* measures.

Most publications are protected by *copyright*, which means that neither the text nor the audio my be used without the permission of the copyright owner(s). For the purposes of voice development this restriction can be overcome by using books whose copyrights have expired, but using "old" books introduces other complicating factors to the voice building process, some of which will be addressed in this paper.

A synthetic voice cannot be built and distributed using a single speaker's voice without his/her *consent*. A possible solution to this restriction is to combine the voices of different speakers to create an average voice model that does not contain biometric information of a specific person, e.g. [7], [8]. However, if basic voice development has not been implemented for a particular language, it is highly unlikely that voice adaptation would be a viable option.

To align the audio and text versions of a book automatically, a *digital* version of the text is required. For many languages electronic books are not readily available. If older books are used to avoid copyright issues, the associated text is rarely available in digital format and printed versions are often difficult to get hold of.[2] In these instances optical character recognition (OCR) is required to convert the printed version of a book to electronic format. Although language-independent OCR systems are available, recognition performance can be enhanced significantly by adding language specific information to the process [9]. If this type of information is not available, it could have a negative impact on the accuracy of the OCR process.

Audiobooks require special *pre-processing*, like automatic segmentation and alignment, to enable voice development, e.g. [10]–[12]. The relevant resources and technologies may not always be available in under-resourced languages.

Although researchers in the field seem to agree on a number of standard practices e.g. [13], [14], it is not immediately apparent exactly how "new", less well-known voices should be *evaluated*. While the concepts of preference testing and Mean-Opinion-Scores (MOS) are not difficult to transfer between languages, the design of Semantically Unpredictable Sentences (SUS) has only been specified for a number of languages [15], [16]. The resources required to construct an SUS test may also not be readily available.

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[1]https://www.gutenberg.org/wiki/Main_Page

[2]Moreover, if different editions of the same book were published, it is also a challenge to find the edition that matches the audio.

In this study synthetic Afrikaans[3] voices built using audiobook data were compared with voices derived from professionally recorded voice artists' speech. The aim of the comparison was to investigate the feasibility of audiobooks as an alternative data source, given the challenges and limitations associated with using audiobook material. Despite numerous efforts, permission could not be obtained to use recently published books for this study. As a consequence, only audiobooks of which the copyrights have expired were used.

Two sets of voices were evaluated: male voices built using very small amounts of both data types (around 3 hours, representing a severely resource constrained scenario) and female voices trained on almost 10 hours of audiobook and professional speech data[4]. The comparison addressed two research questions: firstly, how synthetic voices derived from the speech in audiobooks compare with voices derived from professional recordings in terms of naturalness and intelligibility and secondly how the amount of training data available influences the quality of the generated synthetic speech.

The speech data that was used for voice development is described in the next section, followed by an overview of the voice development process in Section III. Voice evaluation is described in Section IV and results are presented in Section V. Concluding remarks are made in Section VI.

II. Data

Speech data was sourced from *Audiobooks* (of which the copyrights have expired) as well as studio quality speech data recorded by professional voice *Artists*. The duration of the four data sets is summarised in Table I.

TABLE I
TOTAL DURATION [MINUTES] OF THE FOUR DATA SETS USED FOR VOICE DEVELOPMENT

	Male	**Female**
Audiobook	206	593
Artist	180	591

A. Audiobook data

Audiobooks were obtained from a local library for the blind. One of the books is *Loeloeraai* by C.J. Langenhoven, published in 1923. The book was read by a male speaker in October 1958. This recording was chosen because it matches a sub-set of the speech produced by the male voice artist. The other books were all Afrikaans narratives read by the same female speaker between 1962 and 1980.

B. Voice artists data

Loeloeraai was included in the text that was read by the Afrikaans voice artists. The recordings made by the male voice artist were used to build an *Artist* version of the audiobook voice[5].

The professional female voice was built using all the voice artist's data that was available at the time of writing, comprising almost 10 hours of speech data. The text that was read during the recordings was selected to ensure phonetically balanced data with adequate diphone coverage in Afrikaans. The prompts comprised a combination of phonetically rich sentences and paragraphs of fictional as well as and non-fictional text. Narrative text was taken from books on which the copyrights have expired.

III. Voice development overview

A. Pre-processing

Audiobooks are usually made available as .wav or .mp3 files. Older audiobooks are sometimes still stored on analogue cassettes. The analogue recordings used in this study were digitised and noise reduction was performed on the digitised versions using the `denoise` function of `SpeexDSP`[6].

The text version of some of the books that were used to develop the audiobook voices is publicly available in electronic format[7]. The other books were scanned and OCR was performed using `ABBY FineReader`[8] with the Afrikaans language model activated. The resulting text contained almost no errors and very little post-editing was required.

As was mentioned previously, pre-processing is required to convert audio and text data into suitable formats for voice development. Aligning the text and audio is an important pre-processing step. Forced alignment procedures, like the Viterbi algorithm, require short audio segments while audiobook speech files tend to be quite long: a single file sometimes corresponds to a whole book chapter. Various techniques have been proposed to address this challenge, e.g. [11], [17], [18].

Some of the proposed strategies specifically address the fact that accurate alignment systems may not be available. For example, poorly trained grapheme models [11] and generalised phone models [19] have been proposed as possibilities when resources are constrained. Unfortunately the scope of the current study did not allow for any of these techniques to be implemented in Afrikaans. Instead, the Afrikaans version of `Aeneas`[9] was used to automatically align the audiobook text and speech data at sentence level.

B. Voice build

The `Speect` TTS system [20] was used to build the voices using the automatically aligned speech and text. `Speect` is a Hidden Markov Model (HMM) based parametric synthesis system that has been developed specifically within the context of TTS for under-resourced languages.

[3]Afrikaans is one of South Africa's 11 official languages and was chosen as an example of an under-resourced language for the purposes of this study.

[4]Ideally, all the voices in the study should have been either male or female. In this instance the experimental design was determined by the data the authors had access to.

[5]Unlike the volunteer at the library for the blind, the voice artist did not read the entire *Loeloeraai* book, hence the slight difference in duration between the two male data sets.

[6]https://github.com/xiph/speexdsp

[7]https://wikisource.org/wiki/Main_Page

[8]https://www.abbyy.com/en-apac/finereader/

[9]https://github.com/readbeyond/aeneas

IV. Voice evaluation

Four voices were built and subsequently compared in terms of objective measures as well as subjective listening tests.

A. Objective evaluation

Two objective measures were calculated for each voice: the mel-cepstral distance (MCD) and the root-mean-square-error (RMSE) of $\log F_0$. MCD is commonly used to measure the accuracy of the spectral envelope and was calculated as the Euclidean distance between the mel-cepstral coefficients of two samples [21]. RMSE $\log F_0$ quantifies the accuracy of the F_0 contour generated by the model. Since the F_0 is only observed in voiced regions, the RMSE of $\log F_0$ was only calculated for the voiced parts of the speech signals.

A subset of 57 sentences from the *Loeloeraai* text was used to calculate the objective measures for the two male voices. An overlapping set of 42 sentences was identified for the two female voices. The average MCD and RMSE of $\log F_0$ between the natural speech and the synthesised speech for each of these sentences were calculated. Dynamic Time Warping (DTW) was applied to ensure that temporal differences between samples did not influence the distance measures.

B. Subjective evaluation

Subjective evaluations were conducted using formal perceptual listening tests administered via a web interface. The two female voices were evaluated first, followed by an evaluation of the male voices about two weeks later. The listening tests were performed by 20 adult listeners native to South Africa who all speak or understand Afrikaans. The majority of the listeners were not familiar with speech technology or synthetic voices.

A paired comparison test was used to evaluate user preference and a Mean-Opinion-Score (MOS) test to evaluate naturalness. Intelligibility was evaluated by means of a transcription test. Fifty test sentences were synthesized, of which 40 were not included in the training data. The remaining 10 sentences were taken from the training data as examples of natural speech (required to compile the MOS test). Each test item also included a *General comments* section where participants could add remarks.

1) Paired comparison test: Each participant listened to 20 voice samples. The samples were presented in 10 sets of two, one *Audiobook* sample and one *Artist* sample. Listeners were required to indicate their preference for each of the 10 sets. All samples were ordered randomly. The listeners' perception of preference was not restricted to any particular aspects of the speech samples.

2) MOS test: Participants were presented with both natural and synthesised examples of each speaker's voice and had to rate the naturalness of the speech. Twenty utterances were presented to each participant, including 10 examples of each voice. In each set of 10 examples three utterances corresponded to the original voice data (natural speech) and seven were synthesised speech. The rating was performed in terms of a 5-point scale, defined as: 1 - Completely unnatural, 2 - Unnatural, 3 - Slightly natural, 4 - Natural and 5 - Completely natural.

3) Transcription test: Transcription tests require SUS, but no guidelines were available for the formulation of Afrikaans SUS. Afrikaans is closely related to Dutch [22] and the design proposed for Dutch SUS could be therefore be followed to some extent. Unfortunately, the only Afrikaans text corpus of substantial size that is available for research was sourced from government documents and contains mostly government specific lexical items and hardly any monosyllabic words [23]. Some manual intervention was therefore required to obtain SUS that adhered to the guidelines specified in [15].

Participants were requested to listen to each speech sample and to provide a transcription of the audio. Twenty SUS were transcribed per listener, of which 10 random samples corresponded to each of the two voices (*Audiobook* or *Artist*). The listeners were allowed to listen to each sample only once. Although the two evaluations were conducted with a two week interval in between, many participants' browsers "remembered" the first set of transcriptions. Two different sets of SUS (generated according to the same procedure) were therefore used to evaluate the female and the male voices.

V. Results

A. Objective evaluation

The objective distance measures calculated for the *Audiobook* and *Artist* voices are shown in Table II.

TABLE II
Average MCD [dB] and RMSE of $\log F_0$ [cent] values for the Audiobook and Artist voices

	Male		**Female**	
	MCD	RMSE	MCD	RMSE
Audiobook	1.7	206	2.1	223
Artist	1.9	185	2.1	151

The results for the MCD indicate that the male voice trained on the audiobook data is slightly closer to the natural voice than the male voice based on the TTS specific data. However, the difference is negligibly small. The MCDs for the two female voices are exactly the same. These results indicate that, with regard to MCD, the acoustic models derived from the audiobook data are just as close to the original data as the models that were trained on the professional voice data.

In terms of the RMSE of $\log F_0$, both the *Artist* voices are closer to the training data than the corresponding *Audiobook* voices. A possible explanation of this observation is the fact that there is much more variation in the prosody of the audiobook data than in the artist data. The volunteers who read at the library for the blind are free to use their voices as they see fit, while the voice artists were instructed to read mostly in a neutral tone. The variation in the audiobook F_0 values could be more difficult to model accurately than the corresponding voice artist values, hence the better match between the voices and the models for the artists' data.

B. Subjective evaluation

1) Paired comparison test: The vast majority of the participants showed a clear preference for the synthetic voices derived from the professional voice artists' data. Given the comparatively poor quality of the audiobook data and the fact that it was "found" data rather than carefully designed data, this result is not unexpected. Many participants remarked that the audiobook voices sounded old-fashioned, but could not explain exactly which properties of the voices made them sound that way.

2) MOS test: The MOS scores for the male and female voices are shown in Figures 1 and 2 respectively. The results clearly show that the natural examples of the *Artist* voices were perceived as completely natural more often than the natural examples of the *Audiobook* voices. This may be linked to the listeners' perception that the *Audiobook* voices sounded old-fashioned.

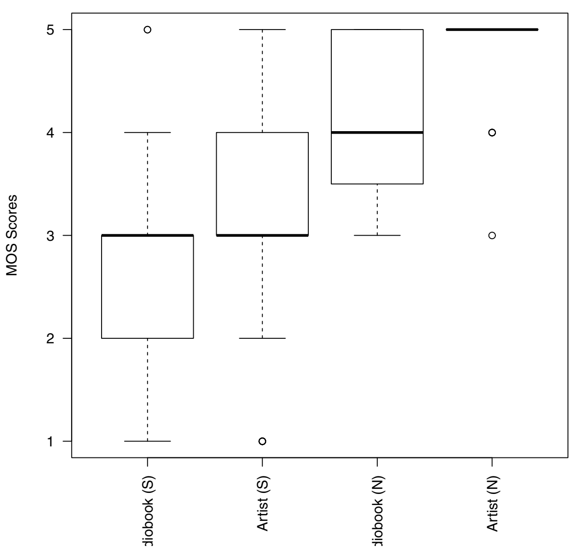

Fig. 1. MOS scores for the *male* synthetic (S) and natural (N) Audiobook and Artist voices.

The data in these two figures seem to indicate that using more training data improves the perceived naturalness of both voice types. While the difference in the perceived naturalness of the male voices is substantial (Fig. 1), the female *Audiobook* voice trained on more data compares favourably with its *Artist* counterpart. However, the *Artist* voices were judged as sounding more natural than the *Audiobook* voices for both the male and the female voices.

3) Transcription test: The average WER for the transcription test was calculated according to guidelines provided in [13]. Spelling mistakes and typographical errors were corrected before the WER values were calculated.

The results in Table III show that using more training data resulted in improved intelligibility for both types of voices. The additional training data makes a more substantial

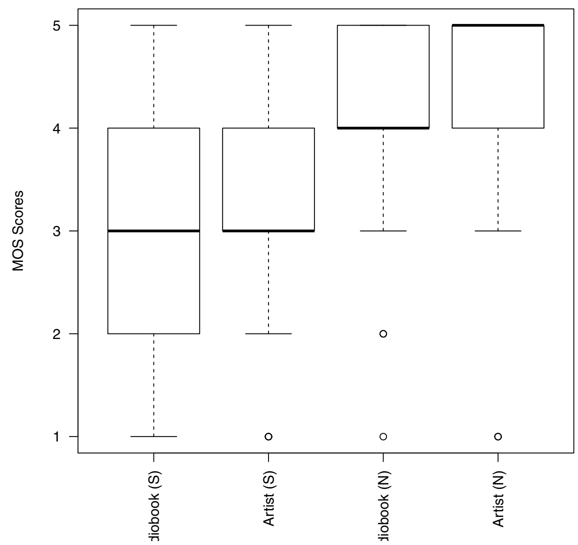

Fig. 2. MOS scores for the *female* synthetic (S) and natural (N) Audiobook and Artist voices.

TABLE III
WER [%] VALUES FOR THE TRANSCRIPTION OF SUS SYNTHESISED WITH AUDIOBOOK AND ARTIST VOICES

	Male	Female
Audiobook	38	25
Artist	19	10

difference to the intelligibility of the *Audiobook* voices than the *Artist* voices. The *Artist* voices are clearly more intelligible than the *Audiobook* voices.

VI. Discussion & Conclusions

The aim of this study was to investigate the feasibility of using audiobooks as an alternative source of data to develop synthetic voices in under-resourced languages. To achieve this aim, two voices built from audiobook material were compared with similar voices derived from professionally recorded voice talent. The results of subjective listening tests indicate that usable synthetic voices can be derived from audiobook data. However, in comparison with professional recordings, much more audiobook speech is required to obtain voices of a similar quality.

Due to copyright restrictions the audiobooks that were used in this study were old and in analogue format. The analogue to digital conversion and denoising operation did not result in acoustic data of the same quality as the studio data. The voice evaluations showed that less data of superior quality yields a better voice - both in terms of naturalness and intelligibility - than much more data of inferior quality. If audiobooks are to be used for voice development, preference should therefore be given to good quality material.

The authors are still trying to obtain permission to use more recently recorded audiobooks to conduct an additional comparative study in the near future. The development of SUS for the country's nine non-Germanic official languages will also be addressed in future work.

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